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→ Second Messenger Mechanism for Mediating Intracellular Hormonal Functions

One of the means by which hormones exert intracellular actions to stimulate formation of second messenger cAMP inside the cell membrane. The cAMP causes subsequent intracellular effects of hormone. Thus the only direct effect that the hormone has on the cell is to activate a single type of membrane receptor. The second messenger does the rest. cAMP is not the only second messenger used by different hormones.

Two especially important ones are

i Calcium ions and associated calmodulin.

ii Products of membrane phospholipid breakdown.

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Table-1.1 Some hormones That uses Adenyl Cyclase-cAMP Second Messenger System

AdrenocorticoTropic hormone (ACTH)
AngioTensin II (epithelial cells)
CalciTonin
Catecholamines (β receptors)
CorticoTropin-releasing hormone (CRH)
Follicle stimulating hormone (FSH)
Glucagon
Human chronic gonadoTropin (HCG)
Luteinizing hormone (LH)
Parathyroid hormone
SecreTin
SomaTostaTin
Thyroid-sTimulaTing hormone (TSH)
Vasopressin (V_2 recepTor, epithelial cells)

→ Adenylyl Cyclase-cAMP

Second Messenger System

Above table shows a few of many hormones that the adenylyl cyclase-cAMP second messenger itself. Binding of the hormones with receptor allows

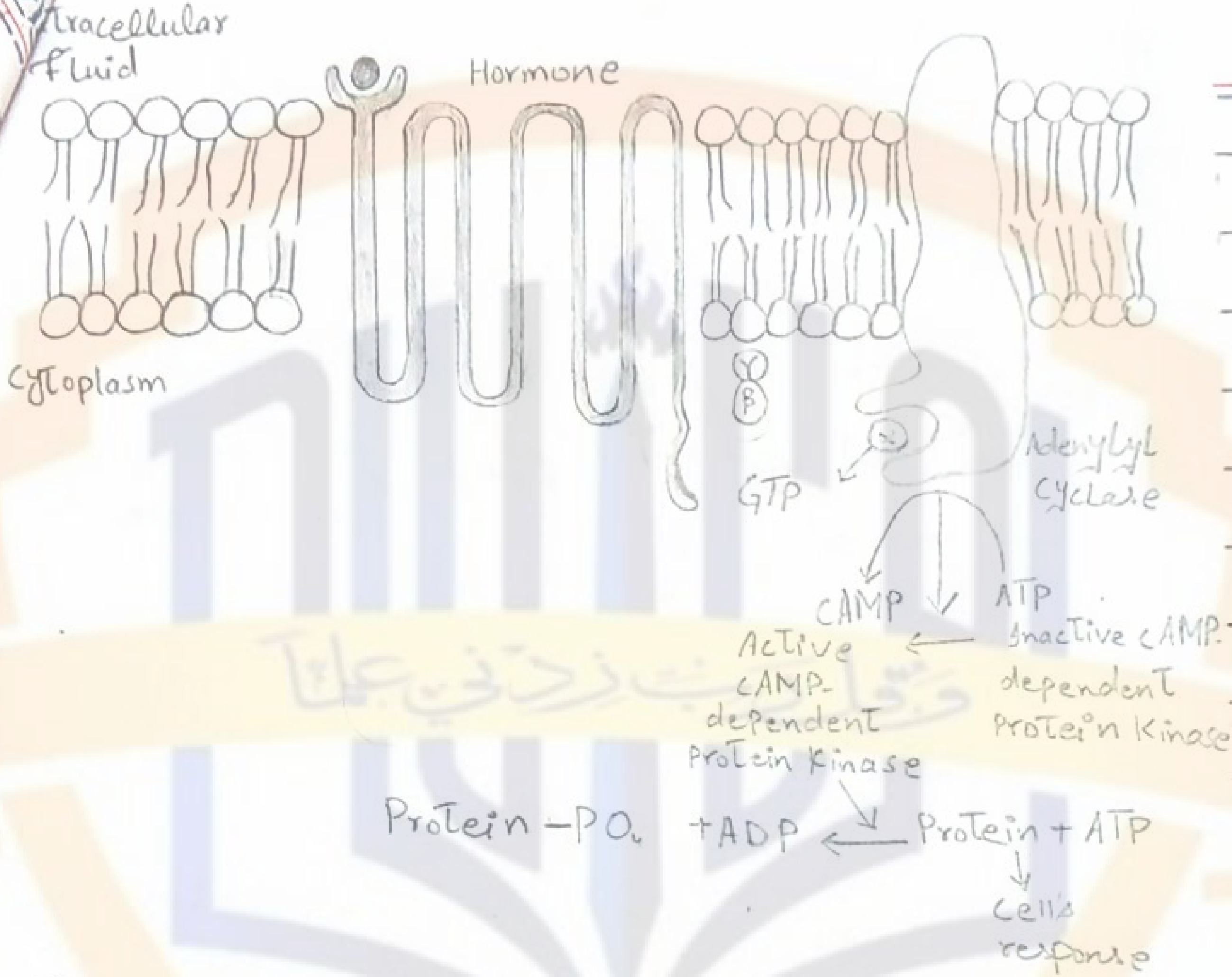

Fig: Cyclic AMP mechanism by which many hormones exert their control of cell function.

coupling of the receptor to a G protein. If the G protein stimulate the adenylyl cyclase-cAMP system it is called a **G Protein.** denoting a stimulatory G protein. Stimulation of adenylyl cyclase a membrane bound enzyme by the G_s protein then catalyzes the conversion of a small amount of cytoplasmic adenosine triphosphate (ATP) into cAMP-dependent protein kinase, which phosphorylates specific proteins in cell, triggering biochemical reactions that ultimately lead to cell's response to the hormone.

Once cAMP is found inside the cell it usually activates a **cascade** of enzymes. That is first one enzyme is activated, which activates a second enzyme, which activates a third one and so forth. The **importance** of mechanism is that only a few molecules of activated adenylyl cyclase immediately inside the cell membrane can cause many more molecules of the next enzyme to be activated, which can still cause more molecules of third enzyme to be activated and so forth. In this way even the slightest amount of hormone can cause a powerful cascading activating force for entire cell.

If binding of hormone to its receptor is coupled to an inhibitory G protein, adenylyl cyclase will be inhibited, reducing the formation of cAMP and ultimately leading to an inhibitory action in the cell. Thus depending on coupling of hormone receptor it can increase or decrease the conc. of cAMP

Different functions are elicited in different target cells, such as initiating synthesis of specific intracellular chemicals, causing muscle contraction or relaxation, initiating secretion by cells, and altering cell permeability.

Thus a **thyroid** cell

stimulated by cAMP forms the metabolic hormones **thyroxine** and **triiodothyronine** whereas the same cAMP in an **adrenocortical cell** causes secretion of the **adrenocortical steroid hormones**. In epithelial cells of **renal tubules**, cAMP increases their permeability to water.

The cell Membrane Phospholipid Second Messenger System

Some hormones activate transmembrane receptors that activate the enzymes **phospholipase C** attached to the inside projections of receptors.

This enzyme catalyzes the breakdown of some **phosophlipids** in cell membrane especially **phosphatidylinositol biphosphate (PIP_2)** into 2 different second messengers products: **inositol triphosphate (IP_3)** and **diacylglycerol (DAG)**. The IP_3 mobilizes calcium ions from mitochondria and the endoplasmic reticulum, and the calcium ions then have their own second messenger effects such as smooth muscles contractions and changes in cell secretions.

DAG the other lipid second messenger activates the enzyme **Protein kinase C (PKC)** which then phosphorylates a large number of proteins leading to cell's response. In addition to these effects the lipid protein of DAG is **arachidonic acid** which is the precursor for the **prostaglandis** and other local hormones the cause multiple effects in tissues throughout the body.

→ Calcium-Calmodulin Second Messenger System

Another second messenger system operates in response to entery of calcium into cells. Calcium entery may be initiated by

(1) Changes in membrane potential that open calcium channels or

a hormone interacting with membrane receptors that open calcium

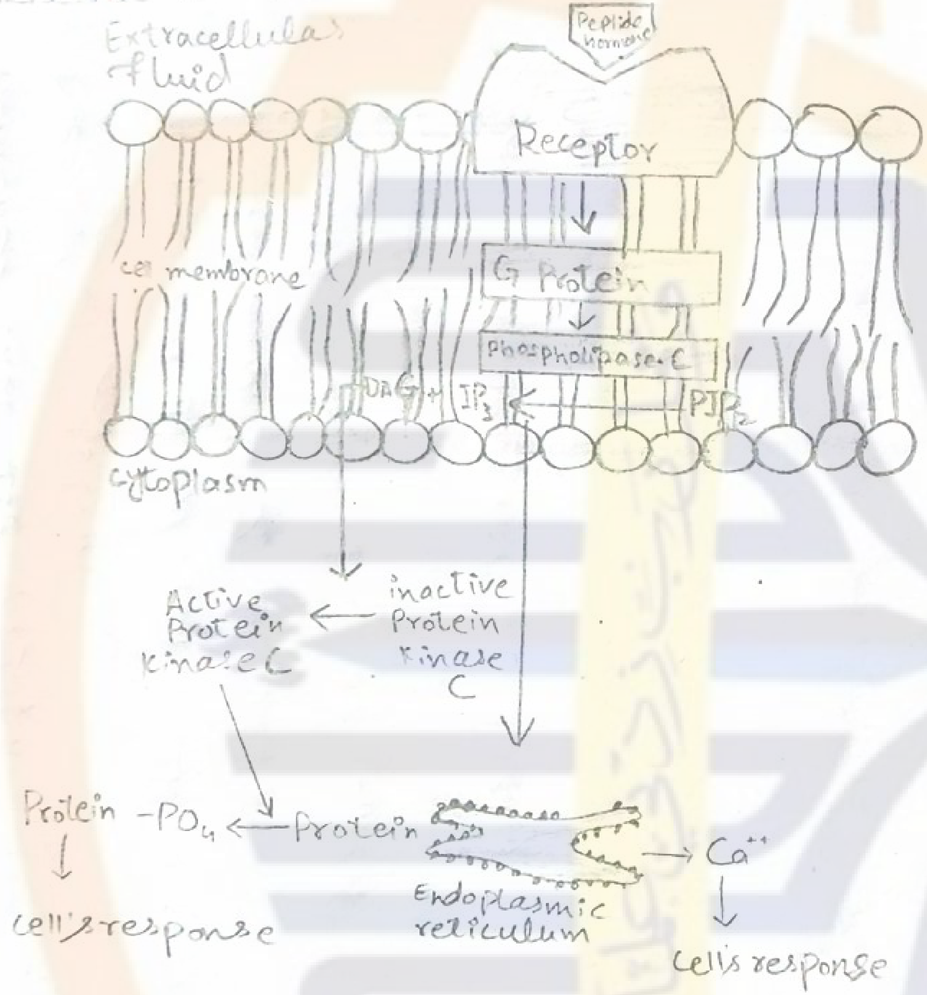

Fig: Cell Membrane Phospholipid Second Messenger System

Table 1.2

Hormones that use Phospholipase C Second Messenger System

Angiotensin II (Vascular smooth muscles)
Catecholamines (α receptors)
Gonadotropin-releasing hormone (GnRH)
Oxytocin
Thyroid releasing hormone (TRH)
Vasopressin

Channels.

On entering a cell calcium ion binds with the protein **calmodulin**. This protein has four calcium sites and when three or four of these sites have bound with calcium, the calmodulin changes its shape and initiates multiple effects inside the cell, including activation or inhibition of protein kinase.

For **example** one specific function of calmodulin is to activate **myosin kinase** which acts directly on myosin of smooth muscles to cause its contraction.

The normal conc. of calcium ion in cells of body is **10^{-8} to 10^{-7} mol/L** which is not enough to activate calmodulin. But when calcium ion conc. rises to **10^{-6} to 10^{-5} mol/L** enough binding occur to cause all the intracellular actions of calmodulin. It is

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interesting that troponin C is similar to calmodulin in both function and protein structure.

Topic # 02

→ Signal Transduction Pathways

→ Intracellular Signaling After Hormone Receptor Activation:

Almost without exception a hormone effects its target tissues by first forming a hormone-receptor complex. This alters the function of receptor itself and the activated receptor initiates the hormonal effects. There are few examples of different types of interactions.

1) Ion Channel-Linked Receptors

Virtually all neurotransmitters substances, such as acetylcholine and norepinephrine combine with receptors in postsynaptic membrane. This almost always causes a change in structure of receptor usually opening or closing of of channel for one or more ions. Some these **ion-channeled-linked receptors** open or close channels for sodium ions, others for Potassium ions, others for calcium ions, and so forth. Their altered movement causes subsequent effects on **postsynaptic cells.** Although a few hormones may exert some of their actions through activation of ion channels receptor most hormones that open or close ions channels do this indirectly by coupling with **G protein-linked** or enzyme linked receptors.

→ G Proteins-Linked Hormone

ii Receptors:

Many hormones activate receptors that indirectly regulate the activity of target receptors proteins by coupling with groups of cell mem-brane proteins called **hetrotrimeric GTP Binding proteins** (G Proteins). There are more than 1000 known G protein coupled

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receptors all of which have **seven** transmembrane segments that loop in and out of cell membrane.

Some parts of receptors that protrude into cell cytoplasm are coupled to G Proteins that include three (**trimeric**) parts — the **α, β** and **γ** subunits. When a hormone binds to extracellular part of receptor a conformational change occur in receptor that activate G proteins and include intracellular signals that either (1) Open or close cell membrane ion channels or (2) Change the activity of an enzyme in cytoplasm of cell.

The **trimeric G proteins** are named for their abilities to bind **guanosine nucleotides.** In their inactive state the **α, β** and **γ** subunits of G proteins form a complex that bind **GDP** on the **α** subunits. When the receptor is activated it undergoes a conformational change that causes the **GDP**

bound trimeric G protein to associate with cytoplasmic part of receptor and to exchange **GDP** for **GTP**. Displacement of GDP by GTP causes the α subunit to dissociate from trimeric complex and to associate with other intracellular signaling proteins, these proteins in turn, alter the activity of ion channels or intracellular enzymes such as **adenylyl cyclase** or phospholipase C which alters cell function.

The signaling event is rapidly terminated when hormone is removed and α subunits inactive itself by converting its bound GTP to GDP.

Some hormones are coupled to inhibitory G protein (G_i) whereas others are coupled with stimulatory proteins (G_s). Thus depending on coupling of a hormone receptor, a hormone can either increase or decrease the activity of intracellular enzymes.

Enzyme-Linked Hormone Receptors:

Some receptors when activated, function directly as enzyme or are closely associated with that they activate. These **enzyme-linked receptors** are proteins that pass through the membrane only once in contrast to the seven-transmembrane G protein-coupled receptors. Enzyme linked receptors have their hormone binding site on outside of cell membrane and their catalytic or enzyme binding site on the inside.

When hormone binds to the extracellular part of receptor, an enzyme immediately inside the cell membrane is activated. Although many enzyme linked receptors have intrinsic enzyme activity, other rely on enzymes that are closely associated with the receptor to produce changes in cell function.

One **example** of an

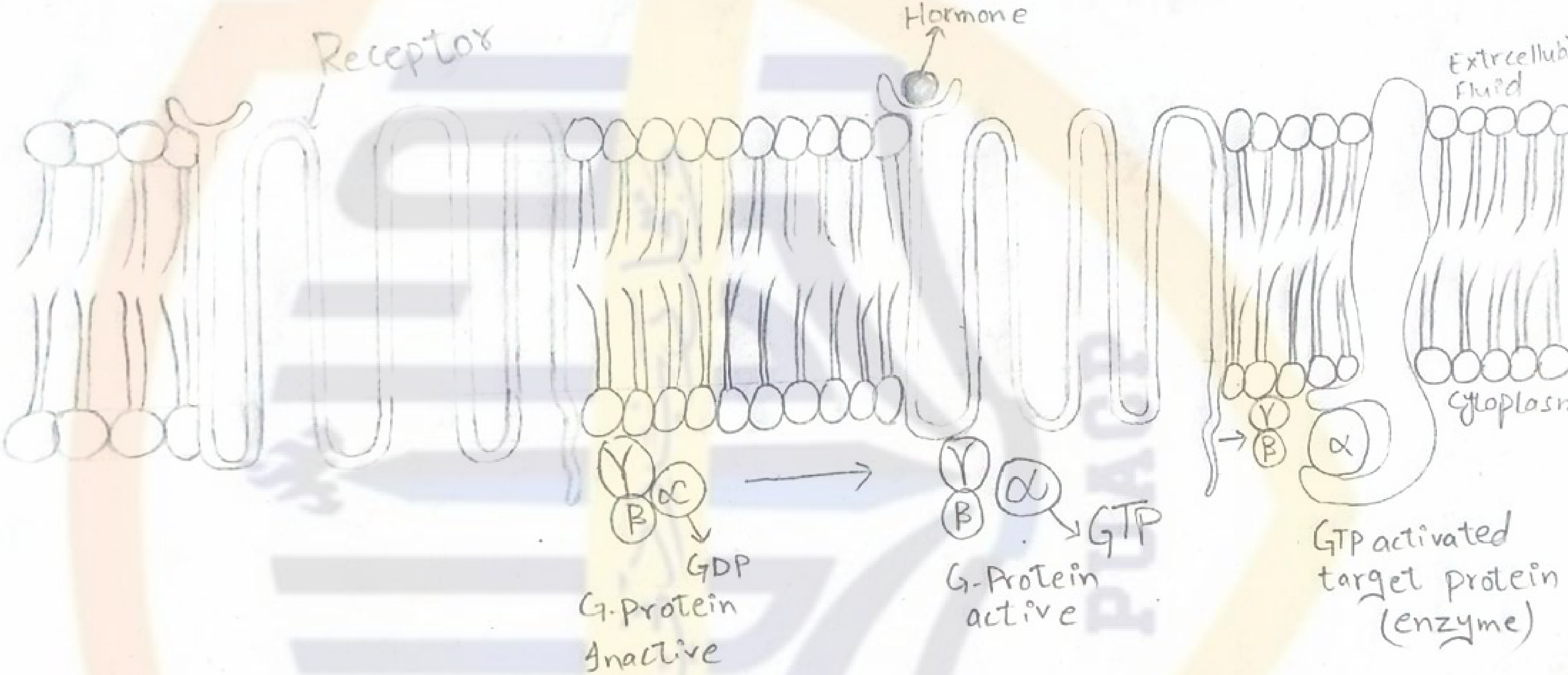

Fig: Mechanism of activation of a G Protein coupled receptor.

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enzyme-linked receptors is the **leptic receptor**. **Leptin** is a hormone secreted by fat cells and has many physiological effects but it is especially important in regulating appetite and energy balance. The leptin receptor is a member of a large family of **cytokine receptors** that do not themselves contain enzymatic activity but signal through associated enzymes.

In the case of leptin receptor one of the signaling pathways occurs through a **tyrosine kinase** of the **janus kinase (JAK)** family, **JAK2**. Phosphorylation of JAK2 also leads to the activation of other intracellular enzyme pathways such as **mitogen activated protein kinase (MAPK)** and **phosphatidylinositol 3-kinase (PI3k)**

An other example, one widely used in hormonal control

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of cell function, is the hormone to bind with a special transmembrane receptor, which then becomes the activated enzyme **adenylyl cyclase** at the end that protrudes to interior of cell. This cyclase catalyzes the formation of **cAMP** which has a multitude of effects inside the cell to control cell activity. cAMP is called **second messenger** because it is not the hormone itself that directly institutes the intracellular changes; instead, the cAMP serves as a 2nd messenger to cause these effects.

iv **Intracellular Hormone Receptors And Activation of Genes**

Several hormones, including **adrenal** and **gonadal steroid** hormones, **thyroid** hormones, **retinoid** hormone and **Vitamin D** bind with protein receptors inside the cell rather than in cell membrane. Because these hormones

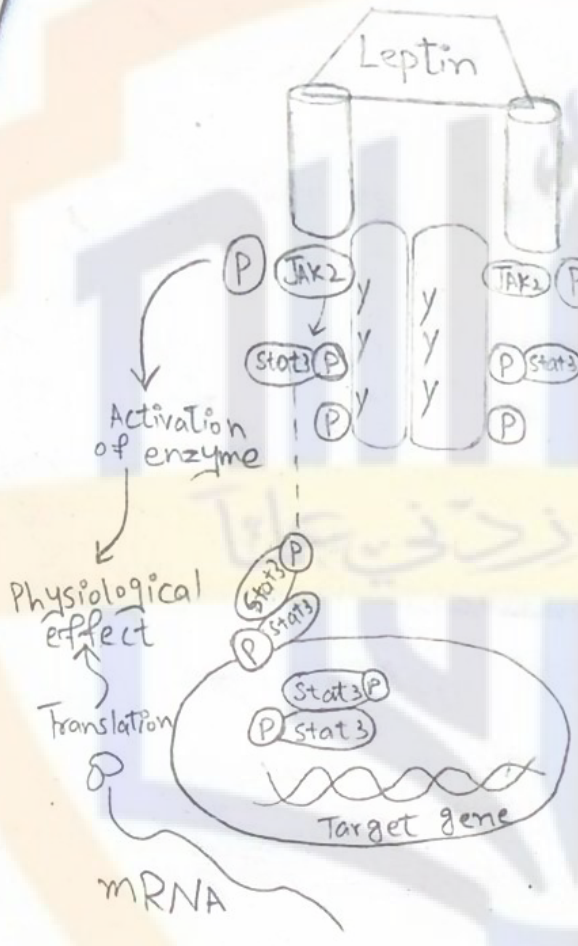

Fig: An enzyme-linked receptor-The Leptin receptor.

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are lipid soluble, they readily cross the cell membrane and interact with receptors in cell membrane or nucleus. The activated hormone receptor complex then binds with a specific regulatory (promoter) sequence of DNA called **hormone response element** and in this manner either activates or respppresses transcription of specific genes and formation of messenger RNA. Therefore minutes, hours or even days after has hormone entered the cell newly formed proteins appear in the cell and become the controllers of new or altered cellular function.

Many different tissues have identical intracellular functions with hormone receptors but genes that the receptor regulate are different in various tissues. The responces of different tissues to a hormone are determined

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not only by the specificity of receptors but also by the genes that the receptor regulates.

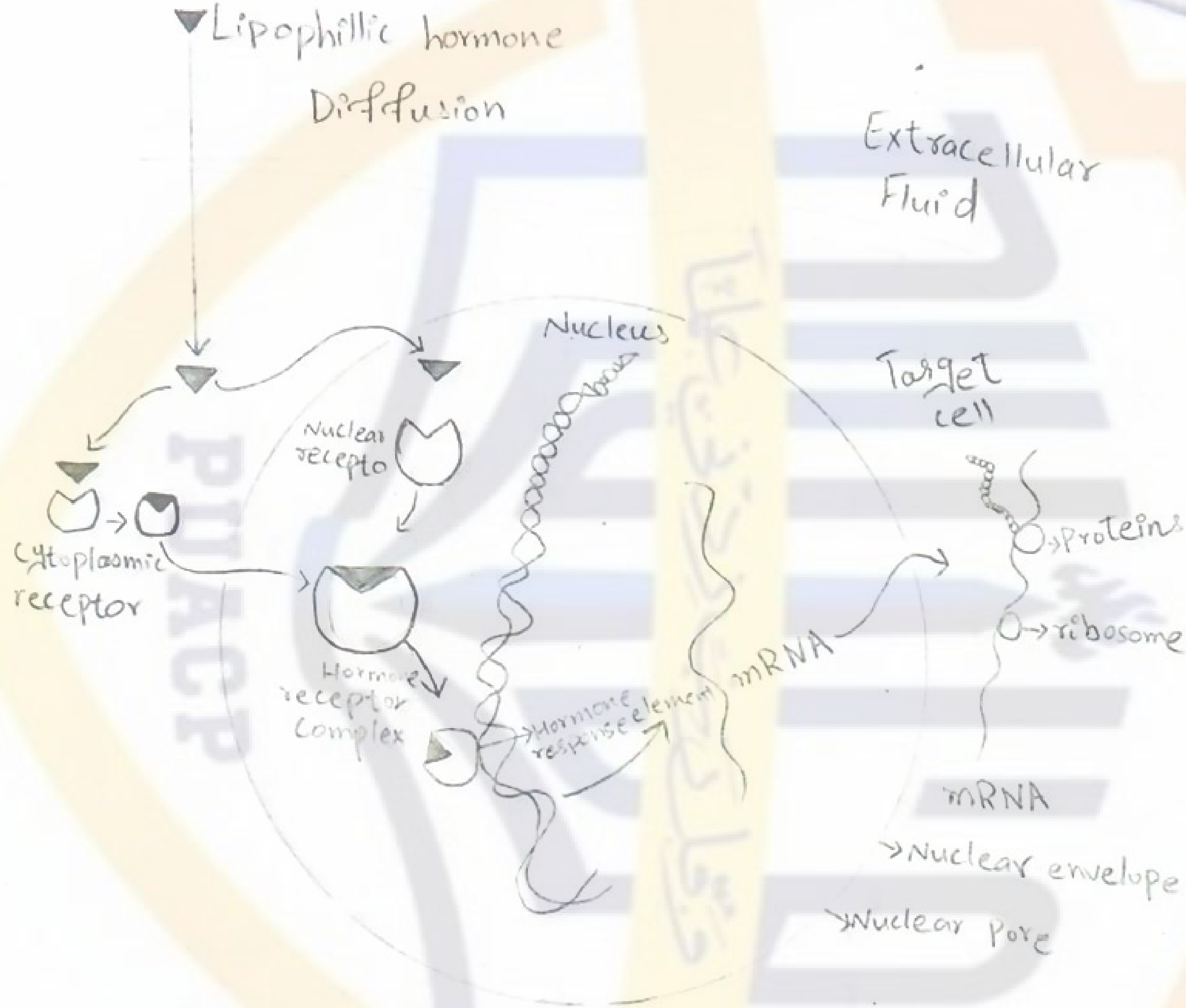

Fig: Mechanism of interaction of lipophillic hormone with intracellular receptors in target cell.